

Q1)

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -1 \\ 1 & 1 & 2 \end{bmatrix}$$

a) Determine the inverse matrix.

Augment the matrix with the identity matrix

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 2 & -1 & 0 & 1 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 \end{array} \right]$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1/2 & 0 & 1/2 & 0 \\ 0 & -1 & 1 & -1 & 0 & 1 \end{array} \right] \quad \begin{array}{l} R'_3 = R_3 - R_1 \\ R'_2 = R_2/2 \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & -1 & 0 \\ 0 & 1 & -1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 1/2 & -1 & 1/2 & 1 \end{array} \right] \quad \begin{array}{l} R'_1 = R_1 - 2R_2 \\ R'_3 = R_3 + R_2 \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & -1 & 0 \\ 0 & 1 & -1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 1 & -2 & 1 & 2 \end{array} \right] \quad R'_3 = 2R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 5 & -3 & -4 \\ 0 & 1 & 0 & -1 & 1 & 1 \\ 0 & 0 & 1 & -2 & 1 & 2 \end{array} \right] \quad \begin{array}{l} R'_1 = R_1 - 2R_3 \\ R'_2 = R_2 + \frac{R_3}{2} \end{array}$$

$$A^{-1} = \begin{bmatrix} 5 & -3 & -4 \\ -1 & 1 & 1 \\ -2 & 1 & 2 \end{bmatrix}$$

b) solution of system of linear equations

System of equation in matrix form

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -1 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$
$$A \quad X = b$$

$$AX = b$$

$$X = A^{-1}b = \begin{bmatrix} 5 & -3 & -4 \\ -1 & 1 & 1 \\ -2 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$X = \begin{bmatrix} 5 - 6 - 12 \\ -1 + 2 + 3 \\ -2 + 2 + 6 \end{bmatrix} = \begin{bmatrix} -13 \\ 4 \\ 6 \end{bmatrix}$$

$$x_1 = -13$$

$$x_2 = 4$$

$$x_3 = 6$$

nullspace of $\begin{bmatrix} -\lambda-3 & 13 \\ -1 & 3-\lambda \end{bmatrix} = \begin{bmatrix} -3+2i & 13 \\ -1 & 3+2i \end{bmatrix}$

$$\begin{bmatrix} 1 & -3-2i \\ -1 & 3+2i \end{bmatrix} \quad R_1' = \frac{-3-2i}{13} R_1$$

$$\begin{bmatrix} 1 & -3-2i \\ 0 & 0 \end{bmatrix} \quad R_2' = R_2 + R_1 \Rightarrow \begin{bmatrix} 1 & -3-2i \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

null space : $x_2 = t \quad x_1 = (3+2i)t$

thus eigenvector

$$v = \begin{bmatrix} 3+2i \\ 1 \end{bmatrix}$$

Q1) a. $0i + 8j + 0k$

b. $0i - \sqrt{5}j + 0k$

c. $0i - \frac{3}{10}j + \frac{2}{5}k$

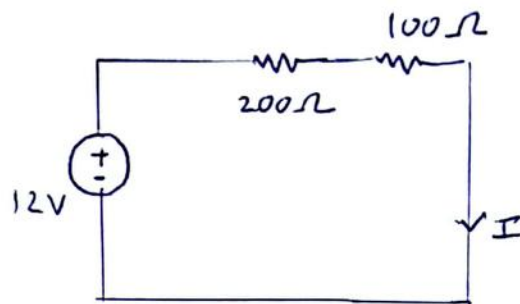
d. $6i - 2j + 3k$

Q2) $r(t) = \left[6 + 3\ln|\sec t| \right] i + \left[5\sin\left(\frac{t}{5}\right) - 7 \right] j + \left[5 - \frac{\ln|\tan 2t + \sec 2t|}{2} \right] k$

Q3) $r(t) = \left[2(t+1)^{3/2} - 2 \right] i + \left[8 - 8e^{-t} \right] j + \left[\ln|t+1| + 1 \right] k$

Q4) $r(t) = \left[2e^t - 3t + 3 \right] i + \left[-e^{-t} + 3t + 6 \right] j + \left[2e^{2t} - 4t + 4 \right] k$

Q1) For steady state circuit is simplified as



using ohm's law $I = \frac{V}{R} = \frac{12}{200+100} = 0.04 \text{ A} = 40 \text{ mA}$

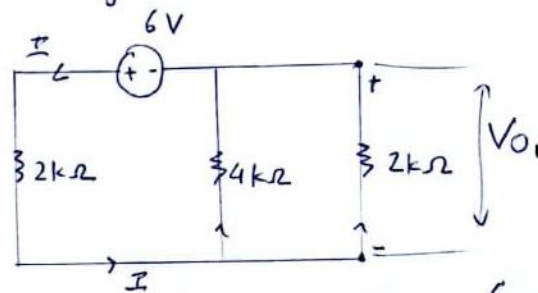
current through inductor = 40 mA

capacitor is in parallel to 200Ω resistor so

$$V_c = 200 \times 0.04 = 8 \text{ V}$$

voltage across capacitor = 8V

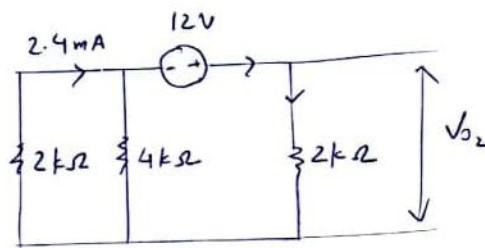
Q3) superposition theorem for V_o
 considering 6V battery first



current through 6V battery: $I = \frac{V}{R_{eq}} = \frac{6}{2 + \frac{1}{\frac{1}{4} + \frac{1}{2}}} = 1.8 \text{ mA}$

$$V_{o1} = -1.8 \left(\frac{1}{\frac{1}{4} + \frac{1}{2}} \right) = -2.4 \text{ V}$$

consider 12V battery now



current through battery: $I = \frac{V}{R_{eq}} = \frac{12}{2 + \frac{1}{\frac{1}{4} + \frac{1}{2}}} = 3.6 \text{ mA}$

$$V_{o2} = 3.6 \times 2 = 7.2 \text{ V}$$

so $V_o = V_{o1} + V_{o2} = 7.2 - 2.4 = 4.8 \text{ V} \Rightarrow \boxed{V_o = 4.8 \text{ V}}$

power supplied by 6V battery

$$P = VI = 6(1.8 - 2.4) \text{ mA} = -3.6 \text{ mW}$$

$\boxed{P_{6V} = -3.6 \text{ mW}}$ so 6V source is charging

Q5)

S.1 : F-value : $\frac{120.60}{10.10} = 11.94$

S. DF = $k-1 = 3-1 = 2$

Adj SS : $12 \times 10.1 = 121.2$

	DF	Adj SS	MS	F value	P value
Factor	2	121.2	120.6	11.94	0.001
Error	12	121.2	10.1		
Total	14	362.4			

S.2: $H_0: \mu_1 = \mu_2 = \mu_3$, $H_1: (H_0 \text{ is false})$

S.3: F value: $11.94 > 1$ so we reject null hypothesis

Here P-value is $0.001 < 0.05$ confidence interval so we reject null hypothesis